

PHYS1007 Lab Report Guidelines

Submission Deadline: Sunday May 12, 2024

General Notes

1. This lab report counts for 15 points (15%) toward your final grade.
2. There are subsections, each subsection will be graded separately.
3. This is an individual project. You can discuss with others and search online resources, but you have to run the experiment and write the lab report by yourself.
4. You are encouraged to use generative AI, such as ChatGPT, to assist with your writing, but grading will emphasise on your logic flow, not on language expression.
5. Submit your report through Canvas before the deadline **Sunday May 12, 2024**.

Your Implementation Task

You need to perform a quantum algorithm on a real quantum computer by using one of the quantum computers in Modern Physics Lab 4117A. The experiment details should be written into a lab report. You can access the quantum computers by booking (https://docs.google.com/document/d/1Q4x6IZE8v-mPav8Jd3u_umou7SvR8eTY_ma4spPtaAg/edit?usp=sharing), in Modern Physics Lab 4117A.

- **Problem description (5 points)**

You need to describe your problem and how to solve it with a quantum algorithm. The problem can be what you have learned in class:

1. Search problem (Grover's algorithm).
2. Quantum Fourier transform
3. Deutsch-Jozsa algorithm
4. Quantum entanglement and Bell state generation

It can also be other quantum algorithms that you are interested in.

If you are not sure about what you choose, feel free to discuss it with TAs.

- **Experiment Implementation(5 points)**

You need to describe how you perform the experiment. The report should contain:

1. The basic information about the quantum computer that you use.
2. How you run the experiment.
3. The experimental result that you obtain.

- **Result Analysis (5 points)**

In this section, you need to give a further analysis of your data:

1. How to interpret your data. What you measure and what it means.
2. How's the real experiment result compared to the ideal case.
3. (Optional) Any possible sources of error?